

Result: $X \sim P_\theta \quad \theta \in \Theta \subseteq \{\pi_k\}$

$$r(\pi_k, T_{\pi_k}) \xrightarrow{k \rightarrow \infty} \sup_{\theta \in \Theta} R(\theta, T^*)$$

$\Rightarrow T^*$ is a minimax estimate

$$\sup_{\theta \in \Theta} R(\theta, T) \geq r(\pi_k, T) \geq r(\pi_k, T_{\pi_k})$$

$$\downarrow$$

$$\sup_{\theta \in \Theta} R(\theta, T^*)$$

Problem: $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Unif}[\underline{a}, \theta]$. & $\theta \geq \underline{a}$

What is ^{a convenient} conjugate family of priors?

$X_{(n)}$ is suff. and

$$X_{(n)} \sim \frac{n x^{n-1}}{\theta^n} \quad \underline{a} \leq x \leq \theta$$

Ans. Pareto Distribution is one choice

$$\left\{ \frac{c(\alpha, a)}{\theta^\alpha} = \pi(\theta) \mid \begin{array}{l} \alpha > 1 \\ \theta > a > 0 \end{array} \right\}$$

Show that any estimate of θ has max. risk ∞

Let $\Theta = \{\theta_1, \theta_0\}$
 $\downarrow$
 f_1, f_0 (pmf/pdf)

$H_0: \theta = \theta_0, H_A: \theta = \theta_1 \quad LR = f_1/f_0$

Test function: (Neyman Pearson Lemma)

$$\phi_{k,r}(x) = \begin{cases} 1 & \text{if } \frac{f_1}{f_0} > k \\ r & \text{if } \frac{f_1}{f_0} = k \\ 0 & \text{if } \frac{f_1}{f_0} < k \end{cases}$$

This test is the most powerful test of its size.

Example: $X \sim \mathcal{N}(\theta, 1)$ $-\infty < \theta < \infty$

$$H_0: \theta = \theta_0 \quad \text{v.s.} \quad H_A: \theta = \theta_1 \\ \theta_1 > \theta_0.$$

Using N-P Lemma, characterize most powerful test.

Most Powerful test has the form

$$\phi(x) = 1 \text{ iff } x > c_\alpha \\ 0 \text{ otherwise}$$

c_α depends on size α .

$$P_{\theta_0}(X > c_\alpha) = \alpha.$$

↳ doesn't involve alternative at all.

∴ The same test works for any $\theta_1 > \theta_0$.

- If instead of simple hyp., we have $\theta \leq \theta_0$ we get uniformly most powerful (UMP) test of its level.

Monotone Likelihood Ratio (MLR) Family.

$$X \sim f_\theta, \theta \in (a, b)$$

If for any $a < \theta_0 < \theta_1 < b$.

$\frac{f_{\theta_1}}{f_{\theta_0}}$ is a monotonic function of T , we call the family MLR.

- C_α became indep of θ_1 since Normal $\in$ MLR family.

Example: $X \sim \text{Bin}(n, \theta)$ $0 < \theta < 1$

Does it have MLR property?

$$\text{LR} \quad \frac{\theta_1^x (1-\theta_1)^{n-x}}{\theta_0^x (1-\theta_0)^{n-x}}$$

MLR w.r.t. X

Q Is it true that Exp family $\subseteq$ MLR family

Ans (Result): The exponential family

$(p(\theta) e^{T(\theta)x} C(\theta) h(x))$ has MLR property

w.r.t. $T(x)$ if C is monotonic in θ .

- ~~For~~ Cauchy distribution $\notin$ MLR family since LR is a ratio of 2 quadratics (taking lim as $\theta_1, \theta_0 \rightarrow \infty$ & as $\theta_1, \theta_0 \rightarrow -\infty$, we get 1)